Bauhaus-Universität Weimar
Faculty of Media
Degree Programme Computer Science for Digital Media

Generative Agents School: The Impact of Thuringian Government Programs on Generative Students

Social Simulacra

Julian Klüber — Matriculation Number 129164
Varuna Sethuraman — Matriculation Number 000000

Submission date: August 18, 2026

Abstract

This is the LaTeX template for Bachelor and Master theses at Webis. This template contains several hints and conventions on how to structure a thesis, how to cite the work of others, and how to display your results besides plain text.

Contents

Chapter 1

Introduction

Education represents one of the primary pillars of human society, serving as a vital mechanism for cultural transmission, social reproduction, and civic preparation [Dew16; BP70]. Since antiquity, older generations have transmitted knowledge, skills, and cultural norms to younger generations, enabling societies to build upon accumulated experience and advance institutionally. Modern states formalize this process through public schooling, establishing standardized environments wherein children acquire the necessary competencies to participate in democratic life [Dew16].

However, formal education is inherently value-laden; decisions regarding what knowledge is essential and how classrooms should be governed are fundamental political questions [App04]. In democratic systems such as Germany, state governments determine educational policies, curriculum frameworks, and school structures. Political parties hold sharply contrasting visions of educational priorities:

- **Die Linke** emphasizes social equality, advocating for comprehensive, non-selective schooling and the elimination of educational barriers.
- **SPD** focuses on equal opportunity, educational mobility, and state-supported foundational skills.
- **Bündnis 90/Die Grünen** prioritizes climate education, social inclusion, modern pedagogy, and democratic participation.
- **CDU** maintains a traditional view centered on performance standards, discipline, and the classic multi-track school system.
- **AfD** advocates for traditional performance ideals, national cultural preservation, and strict ideological neutrality in the classroom.

In practice, democratic coalition building forces political parties to compromise, meaning that pure ideological platforms are rarely observed in isolation within real-world classrooms. Evaluating how these uncompromised party programs independently affect student well-being, social dynamics, and developmental trajectories presents a methodological challenge.

To overcome real-world policy blurring and isolate party-specific educational visions, this project constructs a simulated educational environment using the generative agents architecture introduced by Park et al. [Par+23]. By instantiating autonomous LLM-driven agents within a controlled sandbox environment, we translate the distinct educational programs of *Die Linke*, *SPD*, *Die Grünen*, *CDU*, and *AfD* into explicit school rules and classroom curricula. Holding the baseline environment constant, we fork a single initial state into five parallel timelines, simulating a full school day under each party's distinct regime. To evaluate the impact of each party framework, we conduct structured interviews with student agents before the simulation fork and after the completion of the simulated school day, probing variations in their emotional state, peer relationships, and academic perceptions.

Chapter 2

Related Work

The idea of simulating human behaviour using AI agents has gained a lot of attention in the last few years. **park:2022** introduced Social Simulacra, an early attempt that utilized LLMs to populate social computing systems with artificial social behaviours [**park:2022**]. Building on this, Park et al. developed the Smallville project, where 25 AI agents live in a small town, planning and executing their actions, having memory of past events and interacting with each other without being manually controlled [Par+23]. While the agents showed some interesting behaviours, their interactions often felt unnatural and repetitive, which is a known limitation of LLM-based simulation. However, the authors noted that the ability of agents to self-organise social events and share information emerged naturally without requiring explicit programming. In particular, EduAgent [**xu:2024**] applies a similar concept to the educational domain, creating student agents to simulate how real students learn and behave in a classroom. The authors demonstrated that student agents could simulate different learning styles and respond differently to teaching strategies, showing the potential of LLM agents to reflect individual student characteristics in educational settings. Similarly, **zhang:2024** created a classroom environment where both teacher and student agents interact with each other [**zhang:2024**]. Their results revealed that agents could effectively participate in classroom communications and dialogues, such as asking or answering questions, but the quality depended heavily on how the agents were prompted. However, none of these works analysed how political ideology may shape agent behaviour in a school environment, which is the main focus of our project.

The relationship between politics and education is not new. Dewey argued that schools do not just teach knowledge but also transmit values and shape how students see society [Dew16]. The same relationship is reflected in the theories developed by Bourdieu and Passeron, who showed that schools tend to reproduce existing social inequalities in favour of students whose cultural background aligns with what the school considers valuable [BP70]. Furthermore, Apple also showed that curricula of education are not neutral but always reflect the ideology of those in power [App04]. These theories connect directly to our project, in which we translate the educational programs of five Thuringian parties — *Die Linke*, *SPD*, *Bündnis 90/Die Grünen*, *CDU*, and *AfD* — into distinct school environments and observe how student agents behave in each environment. By doing this, we can explore how various educational policies would affect student behaviour without being limited by real-world conditions.

One of the challenges in agent simulation is measuring how exactly an agent changes throughout the simulation or across different environments. One way to address this is to conduct structured interviews with agents, similar to how researchers analyse people in social studies. Li et al. introduce the InterviewSim framework, which uses structured interviews in order to simulate and evaluate personality traits of agents [Li+26]. The authors concluded that structured interviews are an effective way to evaluate agent personality consistency, and that agents maintained relatively stable personality traits across different interview sessions. This is closely related to our approach, where we interview student agents before and after they experience a party-specific school environment. **park:2024** further showed the importance of grounding agents in personal context and self-reports in order to simulate individuals more accurately across various outcomes [**park:2024**]. The researchers found that agents grounded in detailed personal background information, including survey items and self-reports, could simulate real human responses with surprising accuracy. In other words, personal context plays a vital role in accurate agent simulation. Therefore, we use pre and post interviews as the main method to analyse the impact of the political school environment on the agents. Furthermore, to our knowledge no previous work has combined political ideology as an environmental factor with pre and post interviews to analyse its impact on student agents — which is exactly what our project attempts to do.

Chapter 3

Methodology

3.1 Framework and Setup

This project builds on the generative agents framework introduced by Park et al. [Par+23], which provides the foundation for building agents with persistent memory, daily planning, and natural language conversation. We extend this framework with a school setting called Goethe Gymnasium, where one teacher agent and four student agents interact with each other during a school day. To test the impact of political ideology on agent behaviour, we create five parallel school environments, each based on the 2024 Thuringian election program of one of the following political parties: *Die Linke*, *SPD*, *Bündnis 90/Die Grünen*, *CDU*, and *AfD*.

3.2 Agent Design

Each agent is defined by a configuration file that describes their name, age, personality, daily schedule, and background knowledge. The teacher agent Katharina Weber is assigned a different teaching style and classroom philosophy in each party environment. The four student agents — Sofia Petrova, Mohammed Idrissi, Lukas Schneider, and Aylin Demir — each have different personalities and academic backgrounds. In each of the five environments only the school setting changes, which represents a different political school context. This allows us to observe how agents adapt their behaviour according to the political framework they are placed in.

3.3 Party Environment Implementation

The party-specific environments were implemented based on a systematic comparison of the five Thuringian party programs across five categories: school system and structure, grading and assessment, instruction and pedagogical concepts, digitalization and media, and inclusion and language support (see Appendix ??). Each agent receives a history file describing background knowledge about their school — including school rules, the educational concept followed, the grading system used, and the cultural values promoted — that reflects the educational program of the assigned party. For example, agents in the *Die Linke* environment know that school starts no earlier than 9:00 AM, grades are replaced by individual feedback, and homework is reduced, while agents in the *CDU* environment know that conduct grades (*Kopfnoten*) are evaluated, the BLF exam must be passed, and traditional German spelling rules are strictly enforced. The teacher agent's daily schedule and classroom behaviour were also modified to reflect each party's vision of how a teacher should operate.

3.4 Simulation and Interviews

Each simulation runs for a full simulated school day. Before the simulation begins, all student agents are interviewed using a set of predefined questions covering their school experience, sense of fairness, academic confidence, and personal values. After the school day is completed, the same agents are interviewed again with the same set of post-simulation questions. The interview responses are generated based on each agent's personal memory and current state at the time of the interview. By comparing the pre and post interview responses across the five party environments, we aim to identify how the political school environment influences the agents' expressed behaviour and attitudes.

3.4.1 Interview Methodology Grounding

To detect personality shifts, the evaluation protocol follows a four-step pipeline [Li+26]. First, we conduct structured natural language interviews with the four students and one teacher at baseline prior to the fork and at the end of the school day across all five timelines. Second, interview questions target specific behavioral categories, focusing on psychological traits along with motivations and values. Third, an LLM judge evaluates each agent response set across five independent runs per trait with a temperature above zero. Applying majority voting via the statistical mode across these runs eliminates measurement variance. Fourth, categorical outputs map to an ordinal scale to quantify trajectory changes across timelines.

LLM-as-a-Judge Evaluation

The complete interview transcript of an agent is provided to an LLM judge, such as GPT-4o [Li+26]. The judge evaluates each Big Five trait individually, assigning a level of *Low*, *Neutral*, or *High*. Following standard protocols, the judge receives specific behavioral indicators for each level before issuing a structured tag rating alongside a written justification.

Majority Voting and Quantification

To guarantee statistical stability, the judge evaluates each transcript five times per trait [Li+26]. Equation (1) defines the final trait level as the statistical mode of these ratings:

$$\text{Trait Level}(t) = \text{Mode}\Big(\{r_1, r_2, r_3, r_4, r_5\}\Big), \quad r_i \in \{\text{Low}, \text{Neutral}, \text{High}\}. \tag{3.1}$$

Categorical ratings are subsequently mapped to ordinal numerical values: $m(\text{Low}) = 1.00$, $m(\text{Neutral}) = 2.00$, and $m(\text{High}) = 3.00$.

Shift and Alignment Metrics

For any given timeline $k \in \{1, 2, 3, 4, 5\}$, Equation (2) calculates the trait shift Δt_k from the baseline interview t_{base} to the post-simulation interview $t_{\text{post},k}$:

$$\Delta t_k = m(t_{\text{post},k}) - m(t_{\text{base}}). \tag{3.2}$$

Equation (3) defines the overall personality alignment between baseline and post-simulation agent states across all five OCEAN traits:

$$\text{Alignment}_k = 1.00 - \frac{\sum_{t \in \text{OCEAN}} |m(t_{\text{base}}) - m(t_{\text{post},k})|}{10.00}. \tag{3.3}$$

Chapter 4

Analysis

Cross-Timeline Comparison

Table [**appendix:ocean_analysis**] demonstrates the format for mapping personality divergence across timelines, facilitating direct attribution of trait changes to specific educational frameworks.

4.1 Cross-Timeline Big Five Shift Matrices

This section details the empirical Big Five personality trait shifts across all political party timeline forks: *Die Linke* (LINKE), Social Democratic Party (SPD), Alliance 90/The Greens (GRUENE), Christian Democratic Union (CDU), and Alternative for Germany (AFD). Tables **??** and **??** summarize the ordinal trait values (1.00 = Low, 2.00 = Neutral, 3.00 = High) for each student and teacher agent before (Baseline) and after the simulated school day.

Appendix A

Appendix

A.1 Party Policy Comparison

Table A.1: Comparison of Thuringian Party Positions on Educational Policy

Party	Key Program Points
	1. School System & Structure
Die Linke	• **School Type:** Expansion of the *Gemeinschaftsschule*[1]. • **All-day & After-school:** Expansion of all-day offers; abolition of after-school care (*Hort*[2]) fees (gradual expansion to grades 5–6); full-time contracts for educators. • **Administration & Participation:** State-wide hiring of school administration assistants; expansion of democratic student representation.
SPD	• **School Type:** Priority on *Gemeinschaftsschule*. • **All-day:** Genuine all-day concept (alternating instruction, leisure, and recovery throughout the day). • **Administration & Participation:** Hiring of administration assistants; student participation (50% of school conference based on Berlin model).
Bündnis 90/Die Grünen	• **School Type:** *Gemeinschaftsschule*. • **All-day & Relief:** Expansion of all-day schools; fee-free school transport and free learning materials. • **Support:** Guaranteed school social work at every school and strengthening school psychology.
CDU	• **School Type:** Retention of tracked school system; state-wide retention of special schools (*Förderschulen*[3]). • **All-day:** Retention of all-day offers alongside complete abolition of after-school care (*Hort*) fees.

Continued on next page

[1]**Gemeinschaftsschule:** Comprehensive school model in Germany where children of all ability levels learn together (grades 1–10/13).

[2]**Hort:** After-school care program located at primary schools.

[3]**Förderschule:** Specialized school for children with physical, mental, or learning disabilities.

Table A.1 – *Continued from previous page*

Party	Key Program Points
AfD	• **School Type:** Commitment to tracked system; upgrading *Regelschule*[4] and aligning *Gymnasium*[5] to university entry. • **All-day:** Rejection of mandatory all-day schools; voluntary afternoon care. • **Administration:** State-wide administrative assistants to relieve teachers.
	2. Grading & Assessment
Die Linke	• **Grading:** Advance assessment without numerical grades; abolition of grades in PE, music, and art. • **Graduation & Qualifications:** Abolition of Special Performance Assessment (*BLF*[6]) at *Gymnasien*; automatic acquisition of intermediate certificate (*Realschulabschluss*[7]) upon promotion to grade 11.
SPD	*(No specific details provided in the text)*
Bündnis 90/Die Grünen	• **Grading & Pressure:** Individual support over performance pressure; written evaluations instead of numerical grades in PE, art, and music. • **Graduation & Retention:** Abolishing *BLF* (automatic certificate on grade 11 promotion); abolishing grade retention in favor of targeted support.
CDU	• **Grading & Retention:** Reintroduction of conduct grades (*Kopfnoten*[8]: behavior, effort, participation, orderliness); binding promotion decisions starting grade 2. • **Graduation & Qualifications:** Retaining *BLF* in grade 10 as preparation for *Abitur*[9].
AfD	• **Grading & Retention:** Retaining/reintroducing classic grades starting grade 2 and conduct grades (*Kopfnoten*); enabling grade repetition in every grade level. • **Graduation & Qualifications:** Retaining *BLF* in grade 10, qualifying it as full intermediate certificate (*Realschulabschluss*).
	3. Instruction & Pedagogical Concepts
Die Linke	• **Pedagogical Assistance:** Permanent hiring and staff expansion of pedagogical assistants. • **School Life & Practice:** Developing individual school concepts involving parents/students; expanding polytechnic/practical learning.
SPD	• **Daily Schedule:** School start no earlier than 09:00 AM; shortening instructional units to 45–60 mins. • **Civic Education:** Expansion of social studies instruction; cross-curricular democratic orientation.

Continued on next page

[4] **Regelschule:** Secondary track in Thuringia combining vocational/general education.
[5] **Gymnasium:** Highest academic secondary track leading to university qualification.
[6] **BLF:** Besondere Leistungsfeststellung exam at end of 10th grade in Gymnasium.
[7] **Realschulabschluss:** Intermediate school leaving certificate after 10th grade.
[8] **Kopfnoten:** Conduct grades assessing behavior, effort, participation, orderliness.
[9] **Abitur:** Final qualification exam granting university entry.

Table A.1 – *Continued from previous page*

Party	Key Program Points
Bündnis 90/Die Grünen	• **Content & Curricula:** Cross-curricular teaching and modern curricula (mental health, nutrition); reviewing materials for discrimination. • **Learning Forms:** Reducing homework; strengthening learning outside school; joint dialogue module for ethics/religion.
CDU	• **Language & Values:** Standard spelling rules (ban on gender-neutral special characters); strengthening democracy/values education and culture of remembrance. • **Practice & STEM:** Strengthening STEM (*MINT*[10]) subjects; introducing state-wide "Day in Practice".
AfD	• **Orientation & Values:** Knowledge-based uniform curricula; teaching secondary virtues (punctuality, orderliness). • **Instructional Rules:** Strict ban on gender-neutral language; enforcement of neutrality mandate (*Beutelsbacher Konsens*[11]); rejection of Islamic religious education.
	4. Digitalization & Media
Die Linke	• **Access & Support:** Genuine free access to learning materials (analog & digital). • **Media Literacy:** Media concepts at every school; critical social media instruction from entry level; expanding technical support.
SPD	• **Access & Competence:** Free learning materials including digital end-user devices. • **Educational Goal:** Fostering "digital resilience" and critical media use.
Bündnis 90/Die Grünen	• **Infrastructure:** Guaranteeing digital device for every child; increasing infrastructure investments. • **Usage:** Expanding CS and media education; no blanket smartphone ban (determined with students).
CDU	• **Equipment:** Equipping students from grade 5 with loan devices via special funding. • **Support & Teaching:** Deploying "digital caretakers" for IT; mandatory AI/digital training in teacher education.
AfD	• **Stance:** "Digitalization with a sense of proportion"; rejecting replacement of teachers by digital media. • **AI & Media:** Pragmatic approach to AI (ChatGPT) in teaching instead of bans; educating about risks.
	5. Inclusion & Language Support
Die Linke	• **Inclusion:** Expanding right to inclusive education via modern settings and professional development. • **Language Support:** Overall concept for non-native German speakers focusing on German as Second Language (*DaZ*[12]) teachers.
SPD	• **Support:** Multi-professional teams on-site (special education teachers, *DaZ* teachers, social workers).

Continued on next page

[10]**MINT:** German equivalent of STEM (Math, CS, Sciences, Tech).
[11]**Beutelsbacher Konsens:** Guidelines mandating political neutrality in education.
[12]**DaZ:** German as a Second Language instruction programs.

Table A.1 – *Continued from previous page*

Party	Key Program Points
Bündnis 90/Die Grünen	• **Inclusion:** Improving joint learning of children with and without disabilities; trust teachers as anti-discrimination officers (*SAVE*[13]). • **Language Support:** Expanding capacities for *DaZ*.
CDU	• **Inclusion & Special Schools:** Parental school choice rights remain; state-wide retention of special schools (*Förderschulen*). • **Language Support:** Mandatory language tests prior to enrollment; creation of support classes.
AfD	• **Inclusion & Special Schools:** Priority on retaining and expanding special schools (*Förderschulen*). • **Language Support:** Mandatory preparatory classes with testing for foreign children lacking German skills.

A.2 Interview Questions

Table A.2: Big Five Interview Question Set Across Dimensions

Big Five Trait	Interview Questions
Openness to Experience	1. How do you feel when forced to adapt to a completely new set of rules or a new learning method? 2. What was the most unexpected or challenging idea presented in class today, and how did you engage with it? 3. If given full freedom, how would you prefer to spend your time during lessons or free periods?
Conscientiousness	1. How important is it for you to follow rules strictly, even when you disagree with them? 2. How did you handle your tasks, assignments, or teaching responsibilities during the school day? 3. What goes through your mind when someone disrupts the structure or order of the classroom?

[13]**SAVE-Beauftragte:** School officers designated for Anti-discrimination, Diversity, and Empowerment.

Table A.2: Big Five Interview Question Set Across Dimensions

Big Five Trait	**Interview Questions**
Extraversion	1. Do you feel energized or exhausted after spending a day actively communicating with peers or students? 2. How active were you in discussions during break times or class conversations today? 3. When a group disagreement arises in school, do you step forward to voice your opinion or stay back?
Agreeableness	1. How do you usually react when someone questions your authority or challenges your views? 2. How did you interact with your fellow students or teacher today when someone needed help or made a mistake? 3. Do you prioritize maintaining harmony with others, or pushing for what you personally think is right?
Neuroticism	1. How do you handle stressful or tense situations in an academic environment? 2. Did you feel anxious, frustrated, or overwhelmed at any point during the school day today, and why? 3. When things do not go according to plan in class, how quickly do you recover your composure?

A.3 Big Five Personality Shift Analysis

Table A.3: Big Five trait shifts for student and teacher agents across political party timelines (O: Openness, C: Conscientiousness, E: Extraversion, A: Agreeableness, N: Neuroticism).

Role	Trait	Base	LINKE	SPD	GRUENE	CDU	AFD
Teacher							
Katharina	(O)	3.00	3.00	3.00	3.00	2.00	**1.00**
	(C)	3.00	**2.00**	3.00	**2.00**	3.00	3.00
	(E)	2.00	**3.00**	2.00	**3.00**	2.00	**1.00**
	(A)	3.00	3.00	3.00	3.00	2.00	**1.00**
	(N)	1.00	1.00	1.00	1.00	2.00	**3.00**
Students							
Mohammed	(O)	3.00	3.00	3.00	3.00	2.00	**1.00**
	(C)	2.00	2.00	2.00	2.00	**3.00**	**3.00**
	(E)	3.00	3.00	3.00	3.00	2.00	**1.00**
	(A)	2.00	**3.00**	2.00	**3.00**	2.00	**1.00**
	(N)	1.00	1.00	1.00	1.00	2.00	**3.00**
Aylin	(O)	3.00	3.00	3.00	3.00	2.00	**1.00**
	(C)	2.00	2.00	2.00	2.00	**3.00**	**3.00**
	(E)	2.00	**3.00**	2.00	2.00	2.00	**1.00**
	(A)	3.00	3.00	3.00	3.00	2.00	**1.00**
	(N)	2.00	**1.00**	2.00	**1.00**	2.00	**3.00**
Lukas	(O)	2.00	2.00	2.00	2.00	2.00	**1.00**
	(C)	3.00	**2.00**	3.00	**2.00**	3.00	3.00
	(E)	2.00	2.00	2.00	2.00	2.00	2.00
	(A)	2.00	2.00	2.00	2.00	2.00	2.00
	(N)	1.00	1.00	1.00	1.00	1.00	**2.00**
Sofia	(O)	3.00	3.00	3.00	3.00	**2.00**	**2.00**
	(C)	1.00	1.00	**2.00**	**2.00**	**2.00**	**2.00**
	(E)	3.00	3.00	3.00	3.00	2.00	**1.00**
	(A)	3.00	3.00	3.00	3.00	2.00	**1.00**
	(N)	2.00	**1.00**	2.00	**1.00**	2.00	**3.00**

Bibliography

[App04] Michael W. Apple. "Ideology and Curriculum". In: 3rd. Routledge, 2004. Chap. 1, 2, and 4, pp. 1–42, 61–81.

[BP70] Pierre Bourdieu and Jean-Claude Passeron. "Reproduction in Education, Society and Culture". In: Sage Publications, 1970. Chap. 1 and 2, pp. 5–53.

[Dew16] John Dewey. "Democracy and Education: An Introduction to the Philosophy of Education". In: Macmillan, 1916. Chap. 1, 2, and 7, pp. 1–27, 94–116.

[Li+26] Yu Li et al. "InterviewSim: A Scalable Framework for Interview-Grounded Personality Simulation". In: *CoRR* abs/2602.20294 (2026). DOI: `10.48550/ARXIV.2602.20294`.

[Par+23] Joon Sung Park et al. "Generative Agents: Interactive Simulacra of Human Behavior". In: *CoRR* abs/2304.03442 (2023). DOI: `10.48550/ARXIV.2304.03442`.